

MATH 6359

Optimal Bet Sizing in March Madness using Multivariate Statistics

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Abstract:

Our goal is simple: can we predict the probability that a team will win in the NCAA tournament for each round using the KenPom–BartTorvik dataset? Given a probability of winning, we can compare it to real-world sportsbooks to determine if we have an edge and identify the optimal size of our bankroll to bet on the outcome.

Dataset Creation:

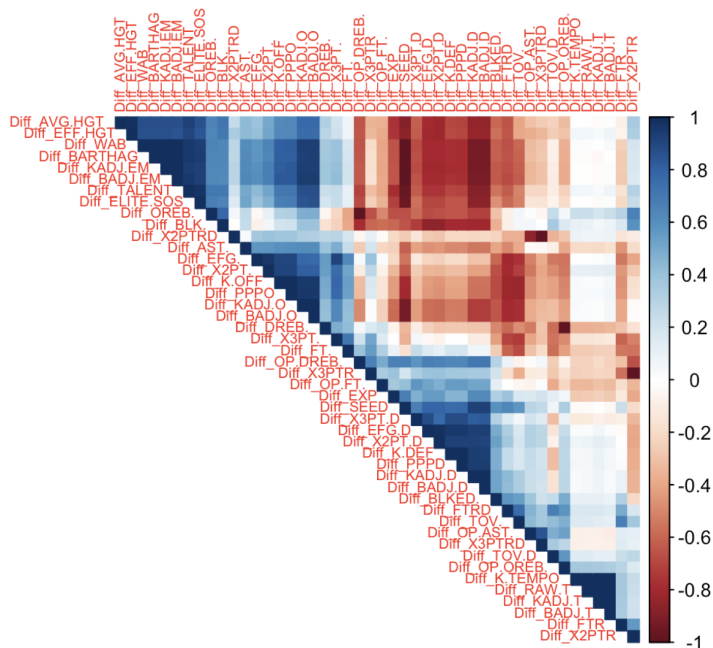
We collected our March Madness dataset from Kaggle which has over 37 datasets. We chose the master data set KenPom_Barttorvik. Ken Pomeroy is a well-known college basketball statistician and has a website <https://kenpom.com/>, with an archive for basketball statistics on every NCAA Division 1 basketball team since the 2002 season. The data set combines it with stat from Bart Torvik which is another prominent basketball analytics expert.

Our strategy is to look at every game from the 2008 to 2025 season and predict the winning team using logistic regression, but first we have over 46 features so we will do the following. Compute the arithmetic difference for every statistic for each matchup, then we will use Lasso Regression as a shortcut to identify the most powerful predictors for Wins. Lasso regression effectively shrinks redundant or weak predictors to zero and we should get 5-8 predictors to use out of the 46. Then we will perform Logistics regression and test on the 2026 march madness bracket and compare outcomes with actual results.

Exploratory Data Analysis:

The data set has 46 features for each team. To better represent the head-to-head matchups, we transformed the data by computing the differences in each teams' statistic between the two teams. This resulted in a new data set where each observation represents a single matchup since 2008. This is to also help to easily eyeball which team is superior in each feature of the data set. This is also better suited to feed our logistic model data since it encapsulates data for both teams in one column rather than treating them independently

To further analyze the dataset, we created a correlation plot for the 46 features of the “Differences in Metrics” from the training data set. The plot reveals multicollinearity among many features and since most features are linear combinations of others. However, we also observe several features that are weakly correlated which may provide more informative signals when putting together our model.



Based on pairwise correlations of the entire dataset of differences between each team metrics, there are over 206 pairwise unique observations where the correlation value of > 0.70 or < -0.70 . So, we acknowledge that there is strong multicollinearity among features and it will be very challenging to sift through all the data manually and analyze their relationships. That brings us to our experimentation process where we have an idea to use LASSO regression to optimize a logistic regression model with the best parameters that don't overfit the data.

Experimentation Process:

Run Lasso Regression of the training set to easily identify strong features

```
```{r}

library(glmnet)

X <- as.matrix(train_data %>% select(starts_with("Diff_")))
y <- train_data$Team_A_Win

lasso <- cv.glmnet(X, y, family = "binomial", alpha = 1)

coef(lasso, s = "lambda.min")

```
```

| | lambda.min |
|---------------|---------------|
| (Intercept) | 1.543167e-01 |
| Diff_BADJ.EM | 6.337868e-02 |
| Diff_FTR | -1.371055e-02 |
| Diff_TOV.D | 1.769504e-02 |
| Diff_OREB. | 8.297945e-03 |
| Diff_OP.DREB. | -6.857166e-16 |
| Diff_AST. | -4.667883e-03 |

Diff_TALENT 2.894049e-03
Diff_WAB 5.236259e-02

Lasso drove all feature of non-importance down to 0 which is not shown here for brevity, but we can see the lambda values range from 0.01543 to -. 000000000000000068571. We removed the lowest lambda value Opponent Defensive Rebounding since it may be highly correlated with Offensive Rebounds. We ran a correlation matrix to check this and can see the perfect correlation between them but interestingly we also see correlations close to 0 for most features. We can see highly positive correlated feature like WAB (Wins above bubble), BarTorvik Adjusted Efficiency Margin, but that is ok since our lambda values for those features show they are significant.

Description: df [8 x 8]

| | Diff_BADJ.EM
<dbl> | Diff_FTR
<dbl> | Diff_TOV.D
<dbl> | Diff_OREB.
<dbl> | Diff_OP.DREB.
<dbl> | Diff_AST.
<dbl> | Diff_TALENT
<dbl> | Diff_WAB
<dbl> |
|---------------|-----------------------|-------------------|---------------------|---------------------|------------------------|--------------------|----------------------|-------------------|
| Diff_BADJ.EM | 1.0000000000 | -0.100721083 | -0.0003813616 | 0.37070112 | -0.37070112 | 0.25958282 | 0.74760146 | 0.940353276 |
| Diff_FTR | -0.1007210828 | 1.0000000000 | -0.0094331652 | 0.20187078 | -0.20187078 | -0.17296876 | -0.02059420 | -0.042646892 |
| Diff_TOV.D | -0.0003813616 | -0.009433165 | 1.0000000000 | 0.12837158 | -0.12837158 | -0.00084210 | -0.09513015 | 0.003438686 |
| Diff_OREB. | 0.3707011188 | 0.201870776 | 0.1283715772 | 1.000000000 | -1.000000000 | 0.07238631 | 0.35267276 | 0.360811013 |
| Diff_OP.DREB. | -0.3707011188 | -0.201870776 | -0.1283715772 | -1.000000000 | 1.000000000 | -0.07238631 | -0.35267276 | -0.360811013 |
| Diff_AST. | 0.2595828177 | -0.172968756 | -0.0008421000 | 0.07238631 | -0.07238631 | 1.000000000 | 0.08975401 | 0.234973763 |
| Diff_TALENT | 0.7476014598 | -0.020594195 | -0.0951301520 | 0.35267276 | -0.35267276 | 0.08975401 | 1.000000000 | 0.715329333 |
| Diff_WAB | 0.9403532762 | -0.042646892 | 0.0034386861 | 0.36081101 | -0.36081101 | 0.23497376 | 0.71532933 | 1.000000000 |

8 rows

Feature Definitions:

- BADJ.EM** - Bar Torvik efficiency margin, this is calculated by taking the # of points scored per 100 possessions (Offensive efficiency) and subtracting the # of points given away on defense per 100 possessions. This captures other correlated features that were dropped like offensive and defensive efficiency
- FTR** – A measure of how often a team gets to the free throw line relative to their field goal attempts
of Free Throw Attempts/Field Goal Attempts
- TOV.D – Defensive turnover percentage.** The % of opponent’s possessions that end in a turnover.
Measure how effective a team is in forcing turnovers
- OREB – Offensive rebound Percentage.** The proportion of a team’s own missed field goals that they successfully rebound. # of Offensive Rebounds/# of
- AST – Assists.** How well the team shares the ball and creates points for the team
- TALENT** – A metric quantifying the raw skill level of a roster, typically calculates suing. Weighted average of recruiting rankings/experience.

7. **WAB** – Wins above Bubble is a resume metric that calculates the number of wins a team has relative to how an average team would perform against that same schedule (Teams in the 40th-50th percentile)

Now that we have a clean dataset and strategy we will perform Multivariate Normality testing to compare the Group of winners to the group of losers.

Multivariate Normality of Features:

```
$multivariate_normality
```

| | Test | Statistic | p.value | Method | MVN |
|---|---------------|-----------|---------|------------|----------|
| 1 | Henze-Zirkler | 1.009 | 0.073 | asymptotic | ✓ Normal |

```
$univariate_normality
```

| | Test | Variable | Statistic | p.value | Normality |
|---|------------------|--------------|-----------|---------|--------------|
| 1 | Anderson-Darling | Diff_BADJ.EM | 2.442 | <0.001 | x Not normal |
| 2 | Anderson-Darling | Diff_FTR | 0.333 | 0.51 | ✓ Normal |
| 3 | Anderson-Darling | Diff_TOV.D | 0.612 | 0.111 | ✓ Normal |
| 4 | Anderson-Darling | Diff_OREB | 0.164 | 0.943 | ✓ Normal |
| 5 | Anderson-Darling | Diff_AST | 0.312 | 0.55 | ✓ Normal |
| 6 | Anderson-Darling | Diff_TALENT | 2.185 | <0.001 | x Not normal |
| 7 | Anderson-Darling | Diff_WAB | 1.830 | <0.001 | x Not normal |

```
$descriptives
```

| | Variable | n | Mean | Std.Dev | Median | Min | Max | 25th | 75th | Skew | Kurtosis |
|---|--------------|------|--------|---------|--------|---------|--------|---------|--------|--------|----------|
| 1 | Diff_BADJ.EM | 1061 | -0.288 | 13.731 | -0.200 | -45.220 | 44.300 | -8.600 | 7.800 | -0.049 | 3.555 |
| 2 | Diff_FTR | 1061 | 0.316 | 6.283 | 0.400 | -24.500 | 23.300 | -3.900 | 4.300 | -0.038 | 3.076 |
| 3 | Diff_TOV.D | 1061 | -0.048 | 3.436 | -0.100 | -13.400 | 11.100 | -2.400 | 2.100 | 0.076 | 3.328 |
| 4 | Diff_OREB | 1061 | 0.082 | 5.706 | 0.000 | -17.800 | 15.900 | -3.700 | 4.000 | -0.033 | 2.844 |
| 5 | Diff_AST | 1061 | -0.572 | 7.359 | -0.700 | -23.900 | 22.100 | -5.500 | 4.000 | 0.078 | 2.904 |
| 6 | Diff_TALENT | 1061 | 1.195 | 42.494 | 1.467 | -94.603 | 93.601 | -28.351 | 31.116 | 0.022 | 2.301 |
| 7 | Diff_WAB | 1061 | -0.239 | 7.286 | -0.100 | -22.700 | 25.700 | -4.800 | 4.100 | 0.039 | 3.535 |

The multivariate normality of the selected features was assessed using the **Henze-Zirkler** test, which produced a p-value of **0.073**, indicating that we fail to reject the null hypothesis of multivariate normality at the 5% alpha level. However, univariate Anderson-Darling tests show that some variables, such as Diff_BADJ.EM, Diff_TALENT, and Diff_WAB, deviate from normality, while others appear approximately normal. This mixed result is very common in real-world scenarios, where individual variables may not be perfectly normal even if the joint distribution is reasonably well-behaved.

MANOVA on **Winning** and **Losing** **Teams:**

```
summary(manova_model)
```

```
***
```

```
              Df  Pillai approx F num Df den Df    Pr(>F)
model_data$Team_A_Win  1 0.26903   55.365    7  1053 < 2.2e-16 ***
Residuals              1059
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We performed a MANOVA to examine whether the combined set of selected features differs between winning and losing teams. The results show a highly significant effect (Pillai's Trace = 0.269, $p < 2.2e-16$), indicating that the multivariate mean vectors of the predictor variables are significantly different across the two teams. This means that the selected features collectively provide meaningful information in distinguishing between winners and losers. Therefore, these variables are appropriate and effective predictors for use in our logistic regression model.

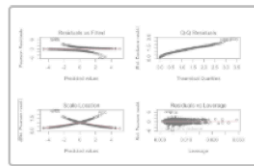
Logistic Regression Model:

```

Start: AIC=1142.64
Team_A_Win ~ Diff_BADJ.EM + Diff_FTR + Diff_TOV.D +
Diff_OREB. +
Diff_AST. + Diff_TALENT + Diff_WAB

Df Sum of Sq  RSS   AIC
Diff_OREB.  1  1126.6 1142.5
Diff_TALENT  1  1126.6 1142.5
Diff_TOV.D   1  1126.6 1142.5
Diff_AST     1  1126.6 1142.5
Diff_FTR     1  1126.6 1142.5
---
R Console

```



Call:

```

glm(formula = Team_A_Win ~ Diff_BADJ.EM + Diff_FTR + Diff_TOV.D +
    Diff_OREB. + Diff_AST. + Diff_TALENT + Diff_WAB, family = "binomial",
    data = train_data)

```

Coefficients:

| | Estimate | Std. Error | z value | Pr(> z) | |
|--------------|-----------|------------|---------|----------|-----|
| (Intercept) | 0.160094 | 0.073337 | 2.183 | 0.029038 | * |
| Diff_BADJ.EM | 0.066646 | 0.017953 | 3.712 | 0.000205 | *** |
| Diff_FTR | -0.031037 | 0.012470 | -2.489 | 0.012811 | * |
| Diff_TOV.D | 0.037631 | 0.021294 | 1.767 | 0.077193 | . |
| Diff_OREB. | 0.019124 | 0.014208 | 1.346 | 0.178292 | |
| Diff_AST. | -0.019443 | 0.010458 | -1.859 | 0.063024 | . |
| Diff_TALENT | 0.003545 | 0.002606 | 1.360 | 0.173728 | |
| Diff_WAB | 0.064395 | 0.030849 | 2.087 | 0.036846 | * |

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

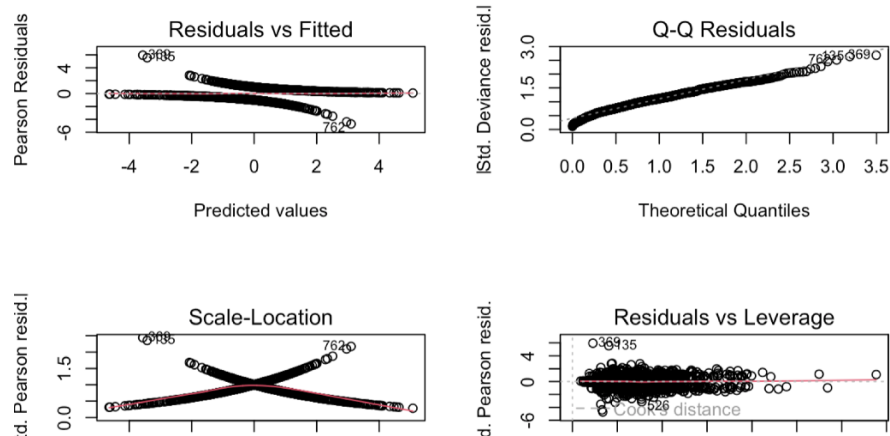
(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1468.4 on 1060 degrees of freedom
 Residual deviance: 1126.6 on 1053 degrees of freedom
 AIC: 1142.6

We fit a logistic regression model using the features selected through LASSO to predict the probability of Team A winning. The results show that **Diff_BADJ.EM** and **Diff_WAB** are strong positive predictors, meaning teams with better efficiency margins and stronger overall resumes are more likely to win.

Diff_FTR is statistically significant with a **negative** coefficient, which suggests that free throw rate differences may reflect differences in play style rather than directly reflecting to wins. **Diff_TOV.D** is marginally significant, indicating that forcing turnovers could still play an important role in determining outcomes.

Overall, the model performs well, as seen by the drop in deviance from 1468.4 to 1126.6, showing a clear improvement over the null model. The AIC value of 1142.6 also suggests a reasonable fit. These results confirm that the selected features provide meaningful insight into predicting March Madness game outcomes.



To assess the validity of our logistic regression model, we examined our 4 standard diagnostic plots. The residuals vs fitted plot shows no strong systematic pattern, suggesting that the model captures the general relationship in the data quite well. The Q-Q plot indicates some deviation from normality in the tails, which is expected in logistic regression since residuals are not required to be normally distributed. The scale-location plot shows mild heteroscedasticity, but not to a degree that significantly impacts the model's reliability. Finally, the residuals vs leverage plot does not reveal any highly influential points that would disproportionately affect the model. Overall, these diagnostics suggest that the model is stable and provides a reasonable fit for predicting game outcomes.

Confusion Matrix for Logistic Regression Model:

Confusion Matrix and Statistics

| Prediction | Reference | |
|------------|------------|------------|
| | Team_B_Win | Team_A_Win |
| Team_B_Win | 28 | 9 |
| Team_A_Win | 5 | 32 |

Accuracy : 0.8108

95% CI : (0.703, 0.8925)

No Information Rate : 0.5541

P-Value [Acc > NIR] : 3.18e-06

Kappa : 0.6216

Mcnemar's Test P-Value : 0.4227

Sensitivity : 0.8485

Specificity : 0.7805

Pos Pred Value : 0.7568

Neg Pred Value : 0.8649

Prevalence : 0.4459

Detection Rate : 0.3784

Detection Prevalence : 0.5000

We evaluated the performance of our logistic regression model. From our confusion matrix and related classification metrics we can see that the model achieved an overall accuracy of **81.08%**, which is significantly higher than the no-information rate of 55.41%. With a p-value of $3.18e-06$, it shows that the model performs substantially better than random guessing.

The sensitivity of 0.8485 shows that the model is effective at correctly identifying games where Team A wins, while the specificity of 0.7805 indicates a strong ability to correctly identify losses. Additionally, the balanced performance between sensitivity and specificity indicates that the model is not heavily biased toward one class. Overall, these results demonstrate that the model provides reliable and meaningful predictions for March Madness game outcomes.

2026 Tournament Prediction:

After validating our logistic regression model, we applied it to the 2026 NCAA tournament matchups to generate predicted win probabilities for each game. For every matchup, the model computes the probability that Team A wins based on the selected features, and the complementary probability is assigned to Team B. The team with a predicted probability greater than 0.50 is selected as the predicted winner.

The results show that the model is able to distinguish between stronger and weaker teams with varying levels of confidence. In many matchups, especially those involving highly ranked teams, the predicted probabilities are significantly skewed toward one team, indicating a clear expected outcome. In contrast, more evenly matched games tend to have probabilities closer to 0.50, reflecting greater uncertainty.

These predictions provide more than just a classification of winners and losers. They offer a probabilistic framework that can be directly compared to sportsbook odds. This allows us to identify situations where the model's estimated probability differs from the market's implied probability, which is critical for determining potential betting opportunities. Overall, the model's ability to generate meaningful and interpretable probabilities makes it a valuable tool for analyzing March Madness matchups.

Optimal Bet Sizing with \$100,000 bankroll:

The key to simulating an outcome for a \$100K bankroll using our prediction probabilities is to gather the Vegas probabilities. We estimated their odds as it would be time intensive research but if we couldn't find odds we set the odds to 5% less than our probability since Vegas odds would be close to ours and they include a “vig” typically +2% or add a cost of doing business to their odds. Their probabilities determine the payout which is important for our calculations.

Here is a sample of our table showing Vegas probabilities side by side and a calculated edge

| GameID | Team_A | Team_B | Pred_Prob | Vegas | Edge |
|---------------|------------------|---------------|------------------|--------------|-------------|
| 1 | Duke | Siena | 0.99 | 0.98 | 0.01 |
| 2 | Ohio St. | TCU | 0.69 | 0.56 | 0.13 |
| 3 | Northern Iowa | St. John's | 0.19 | 0.5 | -0.31 |
| 4 | Cal Baptist | Kansas | 0.19 | 0.5 | -0.31 |
| 5 | South Florida | Louisville | 0.27 | 0.5 | -0.23 |
| 6 | North Dakota St. | Michigan St. | 0.15 | 0.5 | -0.35 |
| 7 | UCF | UCLA | 0.36 | 0.5 | -0.14 |
| 8 | Connecticut | Furman | 0.98 | 0.97 | 0.01 |
| 9 | Lehigh | Florida | 0.02 | 0.5 | -0.48 |
| 10 | Prairie View A&M | Florida | 0.02 | 0.5 | -0.48 |
| 11 | Iowa | Clemson | 0.55 | 0.58 | -0.03 |

Kelly Criterion Example:

Step 1: Convert Odds to "Net Odds" (b)

The Kelly formula uses decimal net odds (b), which represents the **profit per \$1 wagered**.

For American odds of -400:

$$b = \frac{100}{400} = 0.25$$

This means for every \$1 you bet, you stand to make \$0.25 in profit.

Step 2: The Kelly Formula Calculation

We plug your probability ($p = 0.92$) and your loss probability ($q = 1 - p = 0.08$) into the formula:

$$f^* = \frac{bp - q}{b}$$

Substituting our values:

$$f^* = \frac{(0.25 \times 0.92) - 0.08}{0.25}$$

$$f^* = \frac{0.23 - 0.08}{0.25} = \frac{0.15}{0.25} = 0.60$$

Full Kelly Recommendation: 60% of your bankroll.

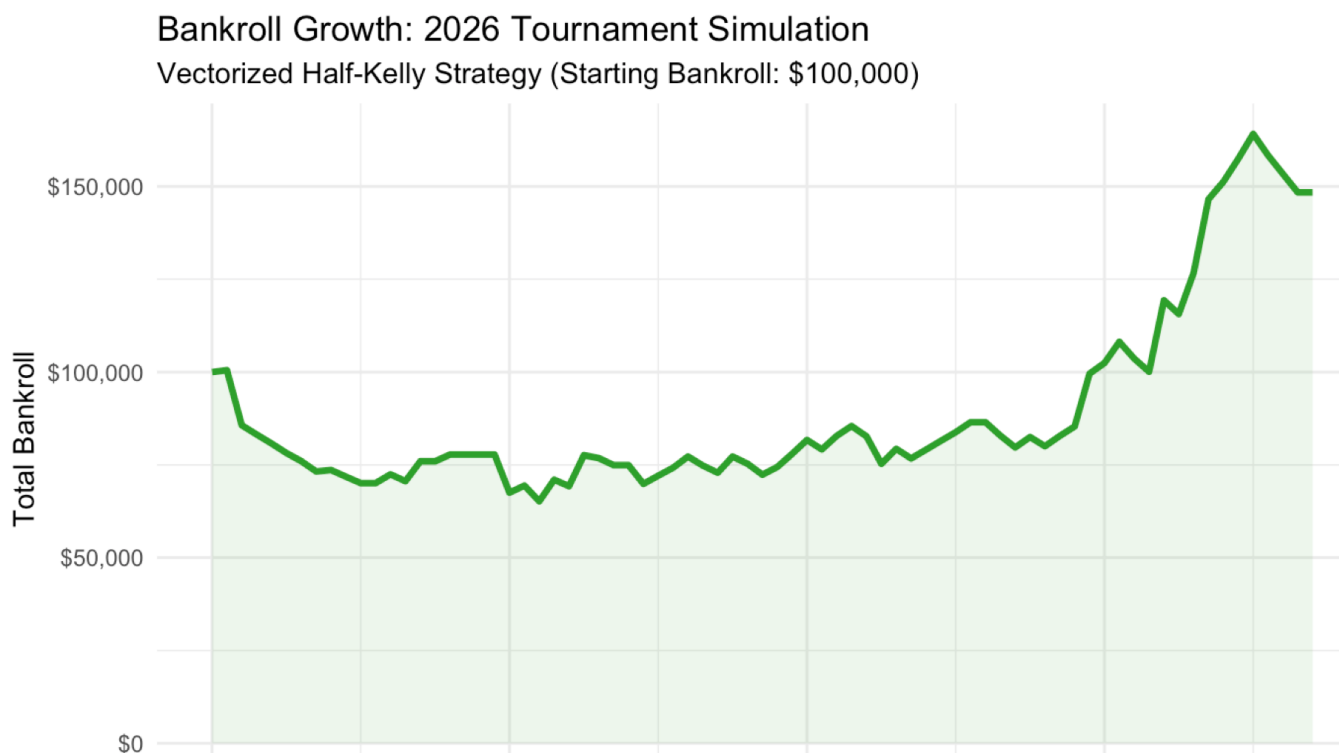
To determine the optimal bet size, we apply the Kelly Criterion, which calculates the fraction of our bankroll to wager based on the model's estimated probability and the betting odds.

For example we can see for American odds of -400 means we win \$0.25 per \$1 we bet

In this example, we assume a model probability of **0.92** and a corresponding loss probability of **0.08**, with betting odds implying a return factor of 0.25. Substituting these values into the Kelly formula, we obtain a recommended bet size of 60% of the bankroll. This result reflects a strong edge, where the model's estimated probability significantly exceeds the implied probability from the odds. In practice, such large bet sizes can be risky, so a more conservative approach, such as using half-Kelly, is often preferred to reduce volatility while still capitalizing on positive expected value opportunities.

We compute this for every game and we are able to simulate a path forward, the optimal bet-size will vary depending on the payout, probability of winning and losing that we generated with our model.

In conclusion for this experiment, this strategy is a wild ride, we see our bankroll draw down below \$75,000 but in the long term we achieve 50% percent returns. We can decrease volatility by using half the suggested bet-size. In some cases our model will show a negative bet size meaning we should not bet based on the inputs provided.



Conclusion:

In this project, we developed a data-driven approach to predict NCAA March Madness game outcomes using advanced basketball analytics. By transforming team statistics into matchup-based differences and applying LASSO regression, we were able to reduce a high-dimensional dataset into a smaller set of meaningful predictors. These features were then used in a logistic regression model, which demonstrated strong performance, achieving an accuracy of over 81% and significantly outperforming a baseline model.

Applying the model to the 2026 tournament allowed us to generate win probabilities for each matchup, providing a more informative perspective than simple win/loss predictions. These probabilities form the foundation for evaluating betting opportunities, as they can be directly compared to sportsbook implied probabilities to identify potential edges. By incorporating the Kelly Criterion, we also established a systematic approach for determining optimal bet sizes, linking statistical modeling with real-world decision-making.

While the model performs well, there are some limitations to consider. The analysis assumes that historical team performance fully captures future outcomes and does not account for factors such as injuries, player form, or in-game dynamics. Additionally, sportsbook odds were not dynamically integrated into the model, which could further refine the betting strategy.

Overall, this project demonstrates that multivariate statistical methods can be effectively applied to sports analytics to generate meaningful predictions and inform betting strategies. The combination of feature selection, probabilistic modeling, and risk-based decision-making provides a comprehensive framework that can be extended and improved with additional data and more advanced modeling techniques.